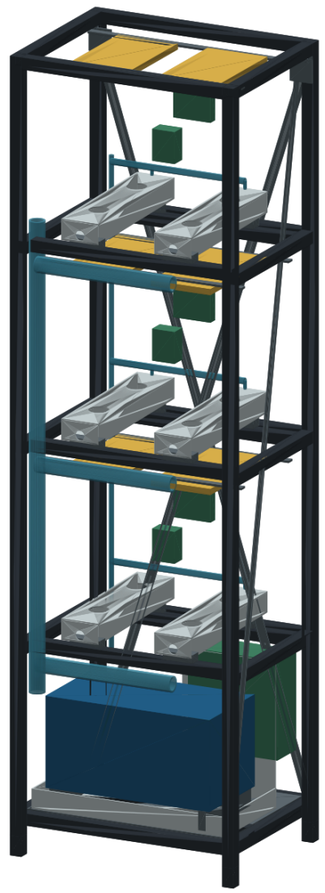
ZOVUYO INNOVATIONS / SMART GROWKIT

Design basis, assembly overview and release status

ZI-SGK-M001

REV A | 15 SEP 2026

Open-frame hydroponic garden



ISOMETRIC REFERENCE / NTS

A dimensioned proposal for engineering review

This package translates the supplied concept into a coordinated mechanical layout. It is not an as-built survey or a signed structural design. Final envelope, tank, bought-out parts and load performance remain open in SSOT v1.0.

Retained source requirements [S]

[S] Provisional maximum 2,000 H x 600 W x 450 D. Three NFT tiers; 12-15 leafy greens/herbs; nominal 20 L tank; ESP32 local control; external certified 12 V supply. See SSOT sections 3, 11 and 14.

This mechanical proposal [P]

12 sites, 6 channels, 30 x 30 modular aluminium frame; each channel 400 long with 8 fall. Four sites per tier. LEDs above every tier. Separate wet plumbing and rear dry electronics bay. Rear X and right-side diagonal bracing. Two wall restraint interfaces.

Reconciled changes

CO2 monitoring is added at your request; sensor model, electrical budget and software schema need updating. No CO2 injection system is included. Parallel feed to each tier is proposed instead of the older concept's cascading return. The source-of-truth NFT baseline takes precedence over the older seedling tray concept.

Limits of the open-frame concept

Fans circulate room air around crops. Temperature, RH and CO2 are measured locally; this open frame cannot independently hold chamber climate or CO2 concentration. An enclosed cabinet would need a separate thermal, ventilation and condensation design.

CAD content and interpretation

STEP: named solids for spatial coordination, with simplified supplier envelopes. DXF: editable full-size geometry and dimensions, one file per sheet; M001 image is PDF-only. PDF: A3 issue set. Hollow extrusion slots, fasteners and flexible fitting details are not modelled.

ENGINEERING REVIEW - NOT RELEASED FOR FABRICATION

All dimensions mm. Do not scale. Proposed dimensions unless marked [S]. HOLD = verify before cutting.

Owner: Oluhle Jawe / Zovuyo Innovations Checked: _____ Approved: _____

Source: SSOT v1.0 + design PDF + video

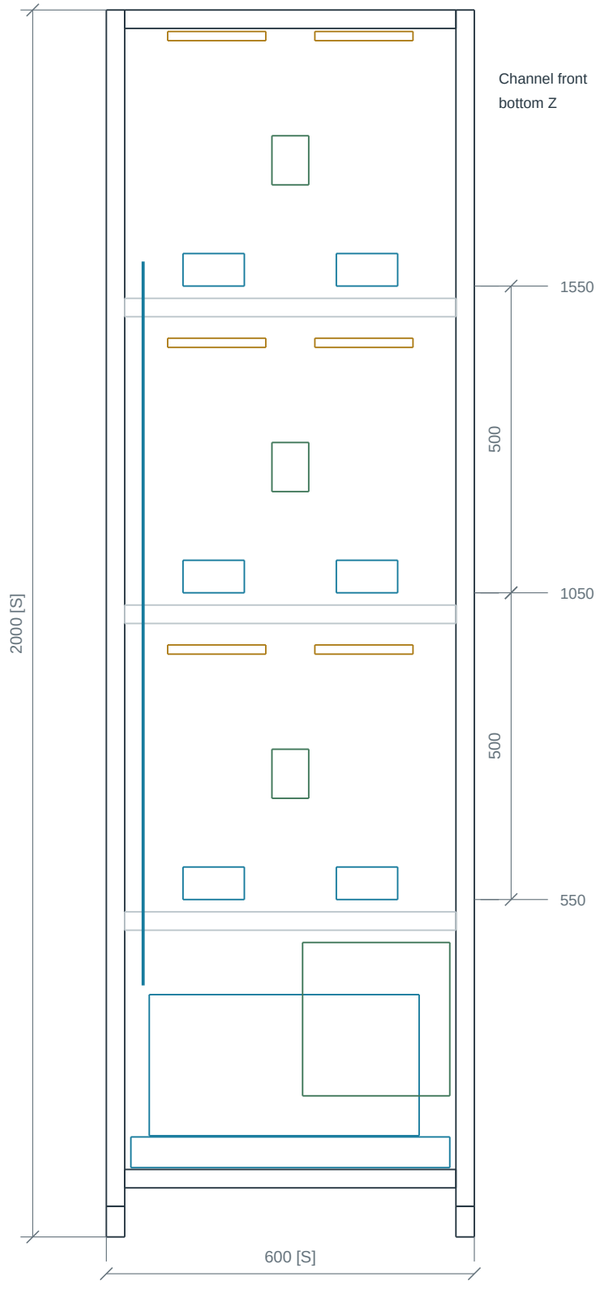
ZOVUYO INNOVATIONS / SMART GROWKIT

General arrangement - front, right and plan views

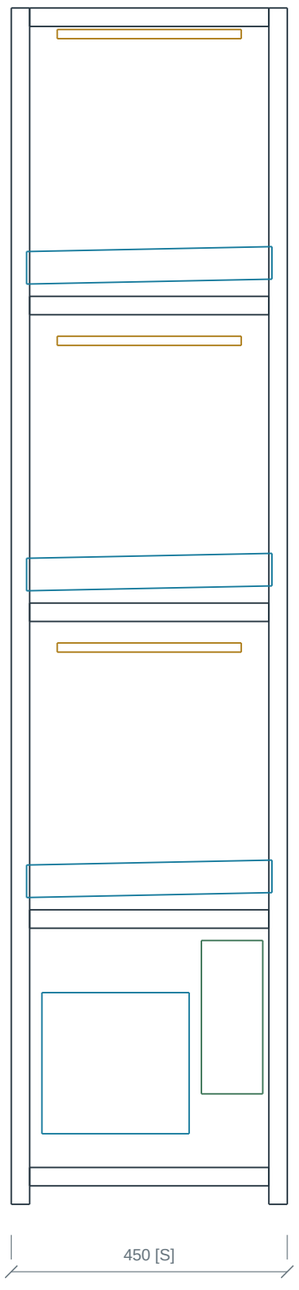
ZI-SGK-M002

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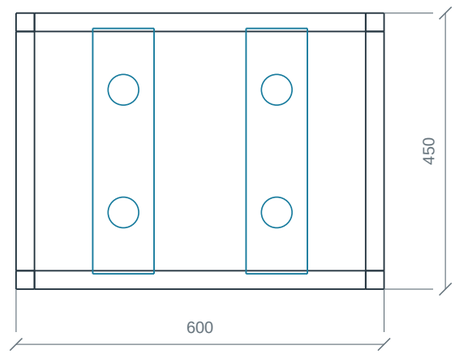
FRONT



RIGHT SIDE



PLAN / LOOKING DOWN



Datums and installation

Origin O = front-left outside corner at finished floor. X to right; Y to rear; Z upward. Front view looks +Y. Right view looks -X. Views individually labelled; no projection-standard symbol implied.

Z=0 is installed floor datum with nominal 50 mm feet. Post ends Z=50 and 2000. Channel front undersides Z=550 / 1050 / 1550. Lid tops Z=603 / 1103 / 1603 at front, +8 at rear.

LED lower faces Z=950 / 1450 / 1950. Minimum rear lid-to-light clearance = 339 mm (plant growth allowance, not guaranteed mature crop clearance). Front user access remains open.

Reserve 600 clear in front for tank servicing, 150 behind for hoses/restraints and 300 on right for electronics service. Proposed installed service zone: 900 W x 1200 D. Confirm user placement before release.

Envelope applies to equipment only, excluding crop canopy, external supply, wall straps and service space. No casters. Relocate only when drained. Foot height adjustment changes overall installed height.

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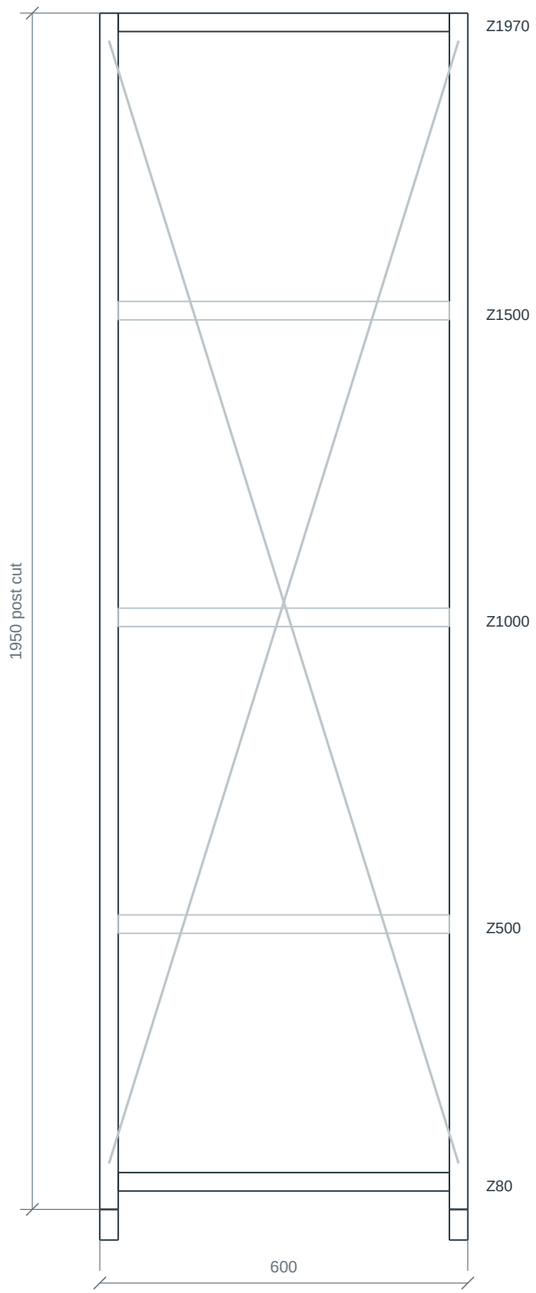
ZOVUYO INNOVATIONS / SMART GROWKIT

Frame fabrication - elevations, members and cut schedule

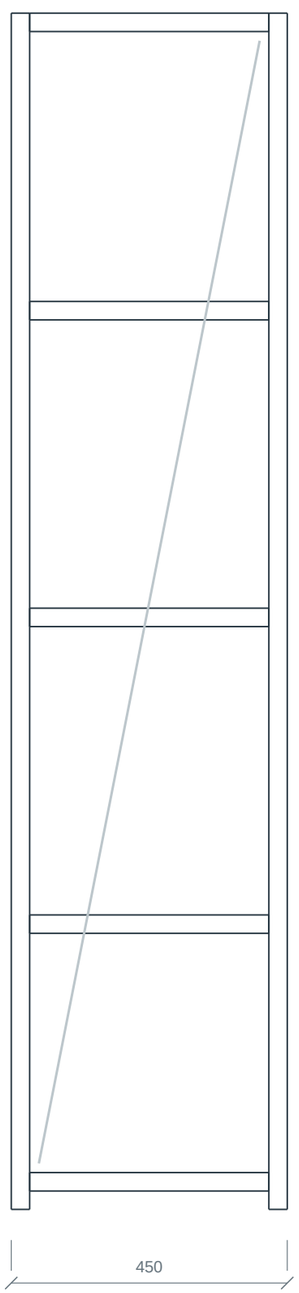
ZI-SGK-M003

REV A | 15 SEP 2026

REAR FRAME



RIGHT FRAME



30 x 30 extrusion cut list - square ends

ID	Member	Qty	Cut mm	Net mm
F01	Posts	4	1950	7800
F02	Width ties	7	540	3780
F03	Side depth rails	10	390	3900
F04	Tier crossmembers	6	540	3240
F05	Base crossmembers	2	540	1080

Total net extrusion: 19,800 mm in 29 pieces. Add saw kerf, end trim and stock optimisation before ordering. F02: two at base, two at top, one rear tie per grow tier. There are no front ties at grow tiers; they would obstruct drains and removal.

Assembly coordinates

Posts: X=0/570, Y=0/420; section 30 square. F02 spans X=30 to 570, at Y=0 or 420. F03 spans Y=30 to 420, at X=0 or 570. Rail underside levels: 80, 500, 1000, 1500, 1970.

F04: X=30 to 570, Y=50 to 80 and 370 to 400 at Z=500 / 1000 / 1500. F05: same X span, Y=140 to 170 and 280 to 310 at Z=80. Base deck top is Z=113.

Joints, bracing and structural hold

Use a single matched extrusion/connector system. Specify at least two orthogonal restraint directions per beam-end joint. Supplier must confirm corner brackets, slot nuts, screws, thread engagement and torque. No universal drilling pattern is authorised.

Rear X brace: centre endpoints (15,125) and (585,1955) in X-Z. Right brace: (45,125) and (405,1955) in Y-Z. See M009. Validate connection stiffness and loaded deflection; section envelopes are not structural calculations.

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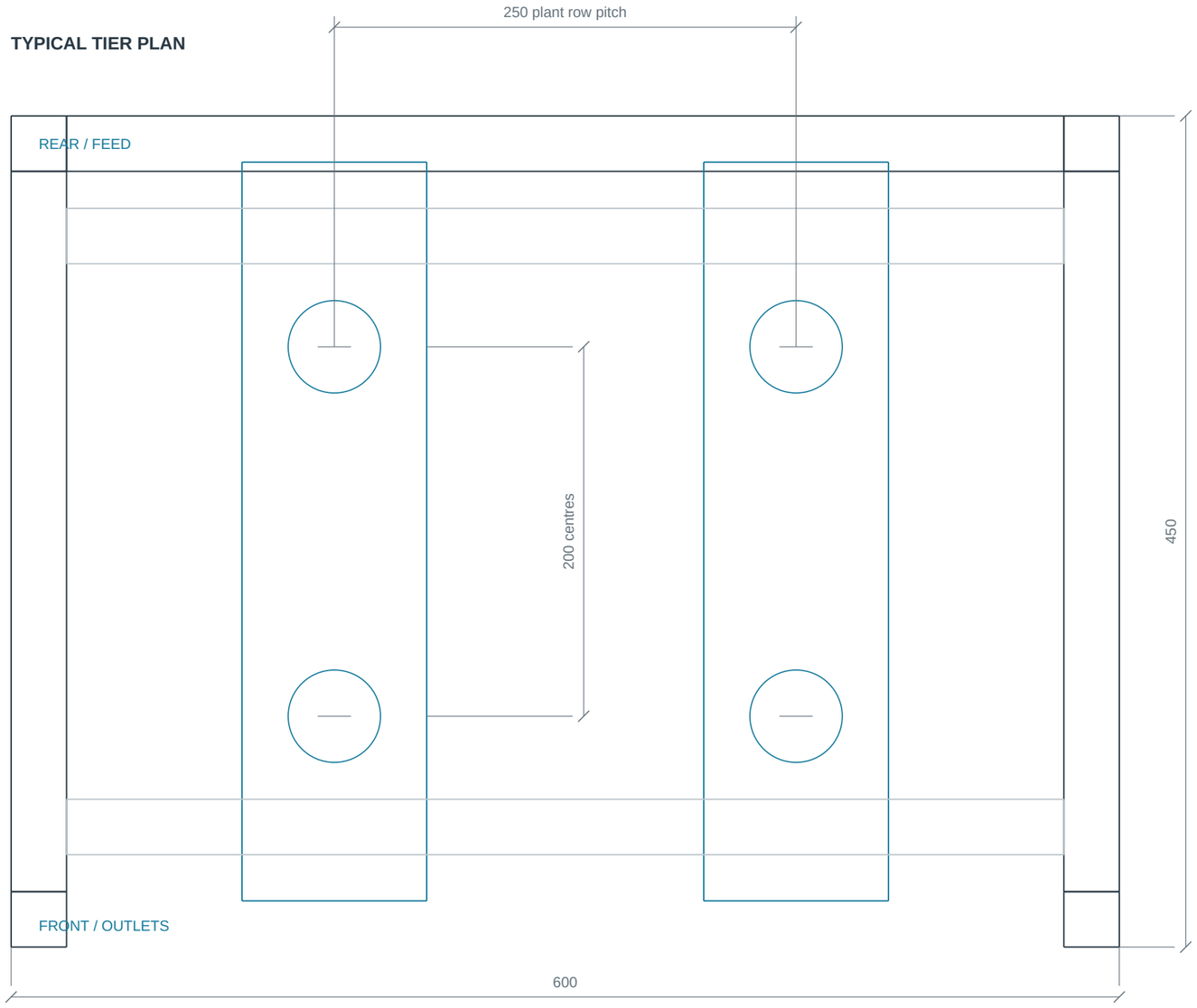
ZOVUYO INNOVATIONS / SMART GROWKIT

Growing tier - plant centres, support and lighting arrangement

ZI-SGK-M004

REV A | 15 SEP 2026

TYPICAL TIER PLAN



Coordinate schedule - all three tiers

Channel footprints: X=125-225 and 375-475; Y=25-425. Plant centres X=175 and 425; Y=125 and 325. Nominal lid holes dia50; final size after net-pot fit test.

100 channel width; 150 clear between channels. Two plant rows are 250 apart; plants within each row are 200 apart. Crop canopy and roots must be verified in a grow trial.

Slope and supports

Channel fall: 8 over 400 = 2.0%, from rear to front. Support centre lines Y=65 and 385, span 320. Channel underside rises 6.4 between these supports.

Rail top is 20 below channel front underside. Required support assembly heights at centres: 20.8 front and 27.2 rear, including pad, plate and adjuster. Use adjustable captive saddles with non-slip isolation pads; match bearing face to 1.146 degree slope.

Lighting and airflow reserves

Two LED envelopes per tier: 160 W x 300 D x 15 H at X=100/340 and Y=75. Two support strips per tier: 540 x 20 x 5, at Y=100/350; underside Z=965/1465/1965. Suspend from the next rail set using adjustable slot hardware.

Fan reserve per tier: 100 W x 25 D x 100 H, X=250, Y=370, Z=780/1280/1780. Add guards on both sides. Select LEDs and fans from measured light coverage, noise, temperature and power budget.

No front width tie at grow tiers; leave the front drain and removal path open.

Two channels per tier x two sites = 4 plants. Six channels total = 12 plants.

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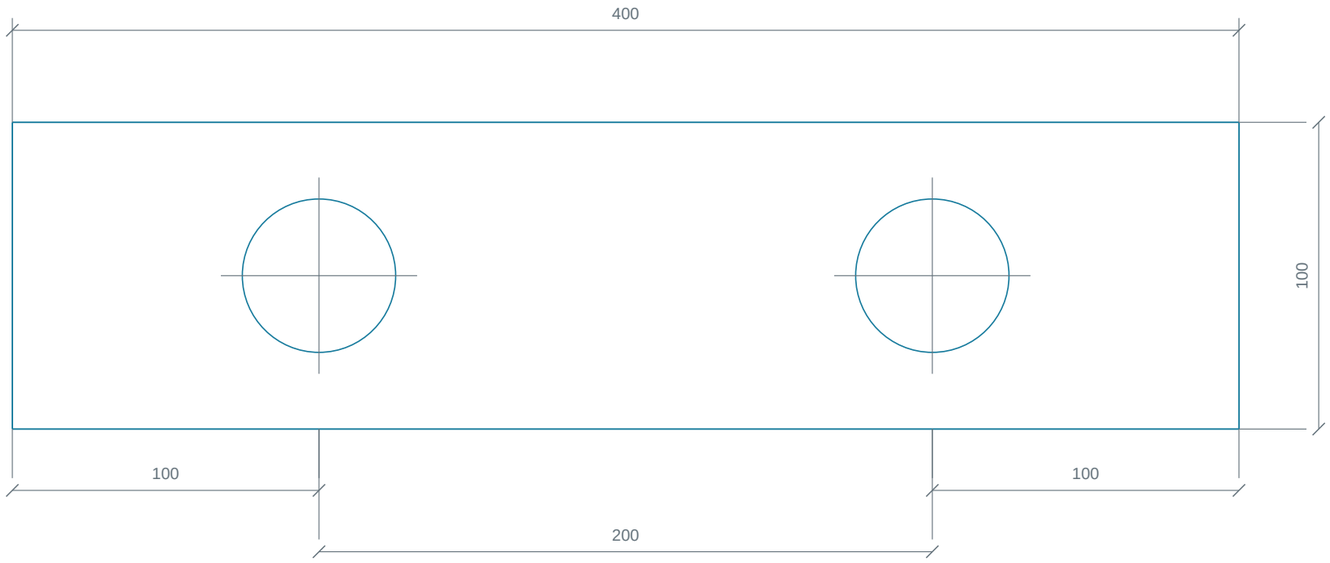
ZOVUYO INNOVATIONS / SMART GROWKIT

NFT channel and removable lid - typical part details

ZI-SGK-M005

REV A | 15 SEP 2026

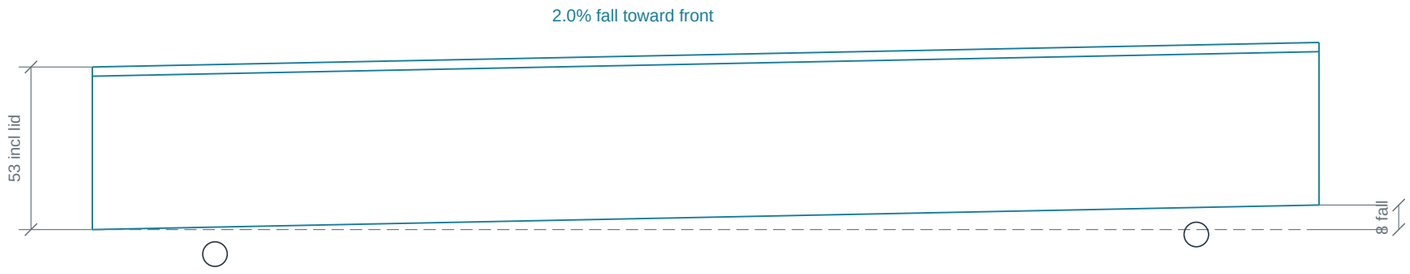
LID PLAN - 6 IDENTICAL



FRONT / LOW END

REAR / HIGH END

SIDE ELEVATION - INSTALLED



Supports at s=40 / 360 along the 400 length; set rise from M004.

H01 channel body / H02 removable lid

Proposed body: 400 long x 100 wide x 50 high, nominal 3 wall/base. Lid: 400 x 100 x 3. These dimensions define the allocated geometry; select a food-contact hydroponic product or validate a fabricated equivalent.

Lid holes: 2 x nominal dia50 through; centres 100 and 300 from the front end, 50 from either long side. Hole size is a HOLD until pot body, lip support and removal are checked. Use removable lids; no adhesive over service edges.

Outlet and feed interfaces - HOLD

Front end outlet reference: centre 50 across width and 17.5 above outside base, nominal dia25 opening. That leaves 2 mm above a 3 mm internal floor. Purchased fitting geometry may change both the hole and centre height. No CNC or drilling release until selected.

Rear feed: 6 ID / 8 OD hose reserve at centre X of channel, Y=405. Keep a visible free discharge above lid; retain nozzle mechanically. Route through a supplier-approved removable entry, not an unsealed arbitrary lid penetration.

Root space, sealing and build limits

Approximate internal width is 94 with the proposed walls. Pot insertion must not bottom out on the 47 nominal internal height. Test pot depth, root mat, outlet blockage and wetting before approving crop use.

Deburr and round plant holes. Use opaque, cleanable, food-contact wetted materials. Channel level across width: within 0.5 mm; installed longitudinal fall: 8 +/-1. End fittings, seals and root guards remain purchased-part details.

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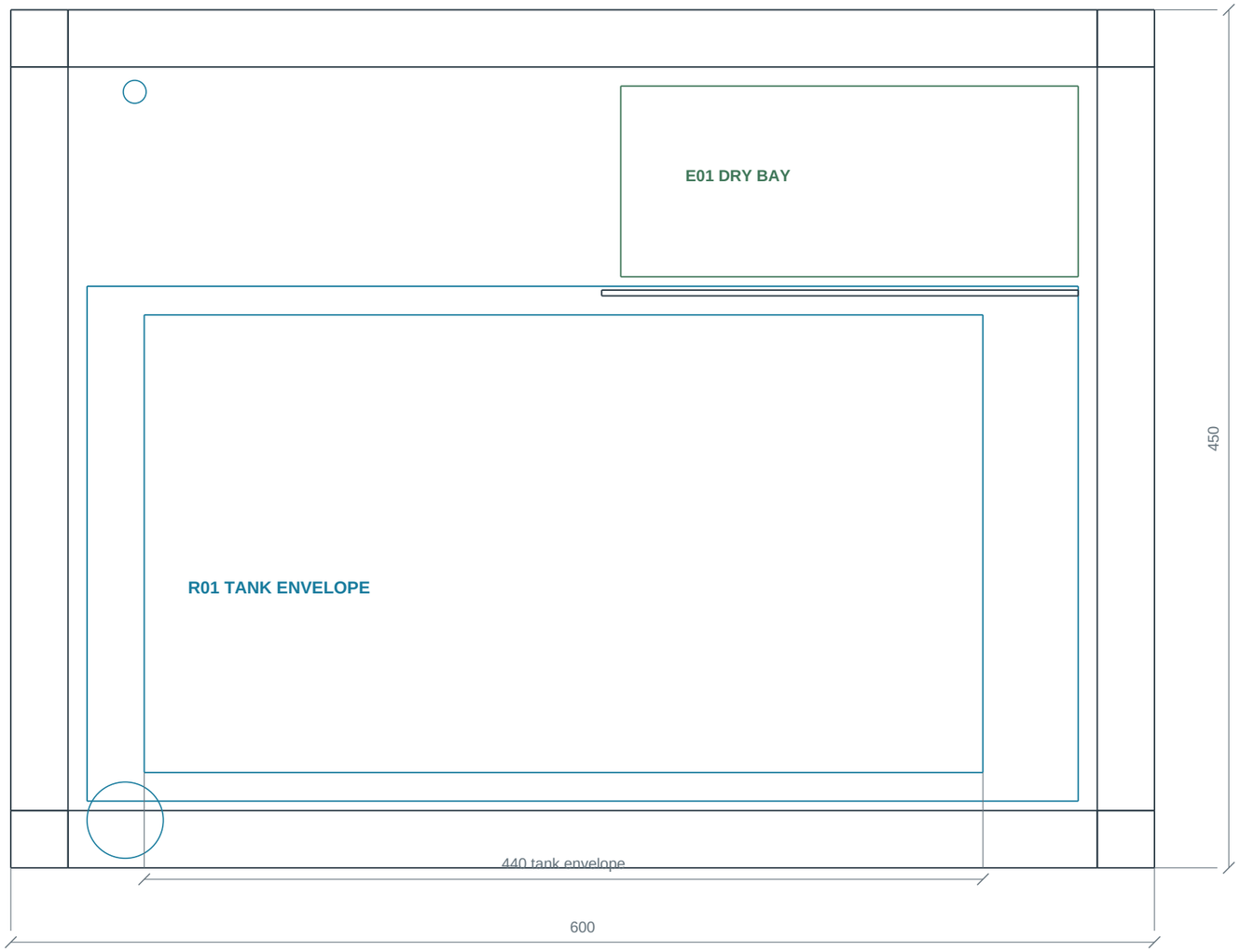
ZOVUYO INNOVATIONS / SMART GROWKIT

Reservoir bay, electronics separation and service access

ZI-SGK-M006

REV A | 15 SEP 2026

BASE PLAN - TANK REMOVES FORWARD



Tank slides/lifts out only after draining and disconnecting hoses and probes.

Spill tray is 520 x 270 x 50 outside; it is not full-reservoir containment.

No permanent fastening through the tank or wet tray base.

Physical envelopes and supports

R01: 440 W x 240 D x 230 H maximum, at X70 / Y50 / Z165. Purchase opaque 20 L nominal tank with removable lid and handles contained in the envelope. Actual tank must be measured before base approval.

Deck B01: 540 x 390 x 3, X30 / Y30 / Z110. Support on perimeter plus two crossmembers. Tank bearers B03: 20 x 240 x 49, X80 / 480, Y50, Z116; non-absorbent, compatible material. Verify vendor tank base supports.

Spill tray B02: 520 x 270 x 50, 3 nominal walls/base, X40 / Y35 / Z113. Tank bottom Z165 is 2 above tray rim Z163. Tray theoretical empty capacity about 6.38 L; bearers reduce it. Leak sensor in lowest available point.

Electronics and probe access

E01 reserve: 240 W x 100 D x 250 H at X320 / Y310 / Z230. Splash barrier E02: 250 W x 367 H x 3 at X310 / Y300 / Z113. Seal barrier joints; maintain removable right-side electronics access.

Keep pH, EC, temperature and level hardware removable from tank lid. Reserve 100 mm above lid for probe handling, then remove drained tank for full cleaning. Probe immersion and holes require selected parts.

Water volume and drain-back

20 L is nominal capacity, not the normal fill mark. Start commissioning at 14 L; increase only after pump-off drain-back test. Proposed maximum operating fill 16 L is a HOLD. Mark actual low, refill and maximum levels after calibration.

Measure all return water and any feed siphoning. Require spare tank capacity >= measured drain-back + 1 L test margin. Low-level cutoff must exceed pump minimum submergence under flow.

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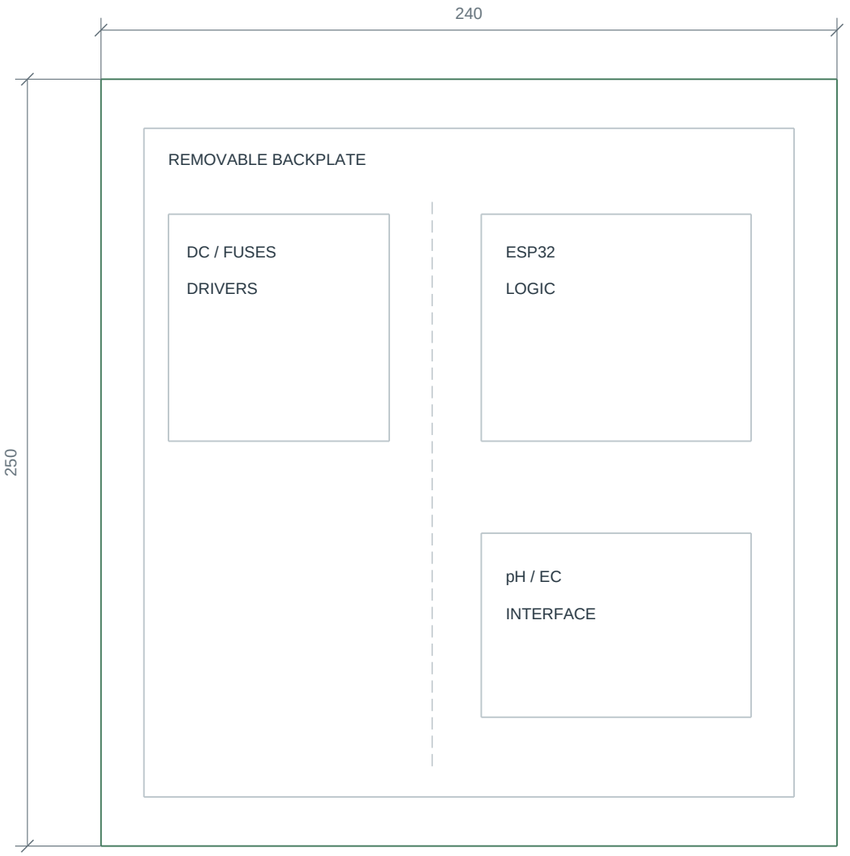
ZOVUYO INNOVATIONS / SMART GROWKIT

IoT mounting interfaces, cable separation and sensing locations

ZI-SGK-M007

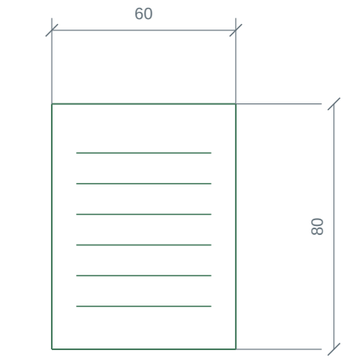
REV A | 15 SEP 2026

E01 REAR BAY ENVELOPE



Internal zones schematic only. No board-hole or gland pattern release.

AIR SENSOR RESERVE



60 W x 40 D x 80 H

Vented and splash-shielded

Sensor schedule and mounting reserves

S01 air temperature/RH: middle pod X270 / Y220 / Z1215. Lower and upper pods at Z715 / 1715 are spare sensing reserves. Mount away from direct LED heat, water spray and direct fan jet; compare readings across tiers.

S06 CO2: add a true CO2 sensor with a ventilated sampling path, preferably NDIR. Reserve within middle pod; final module may require pod enlargement. Do not describe a VOC-derived eCO2 value as measured CO2.

S02 water temperature, S03 pH, S04 EC and S05 level: removable tank-lid probe rail. Respect probe immersion and spacing from each vendor. Use protected level sensing plus independently tested pump-off response.

Leak detection is an added proposal. Air velocity sensing is not included: fan command/tach feedback is not an air-quality measurement. Add particulate/VOC/airflow sensors only through a defined requirement.

Harness and mechanical provisions

Route wet plumbing left. Route protected DC harness on right post in separate removable clips. Keep analogue probe leads away from PWM/load cables. Preserve connector access and minimum bend radii from vendor data.

Bottom-facing glands and drip loops; dedicated sealing washer for every penetration. Route into the enclosure without going through the splash tray. Gland bore and cable clamp size are HOLD until actual cable outside diameters are known.

Use a sealed purchased enclosure suitable for condensation/splash exposure; verify thermal performance. Do not add the older summary's open fan/weep holes. Such holes invalidate an assumed sealed rating.

No mains voltage inside the frame. External certified 12 V adapter only. Final current budget, branch fuses, conductors and connectors follow SSOT section 11; old 10 A master fuse is not a released specification.

User feedback and control

Local ESP32 reads sensors and runs pump, light and fan rules. MQTT/Node-RED links to the PWA for readings, history, settings and alerts. Loss of internet must not stop local operation. Remote commands remain subordinate to low-water and fault interlocks.

CO2 units, calibration metadata and optional sensor availability need a controlled schema update to SSOT v1.1.

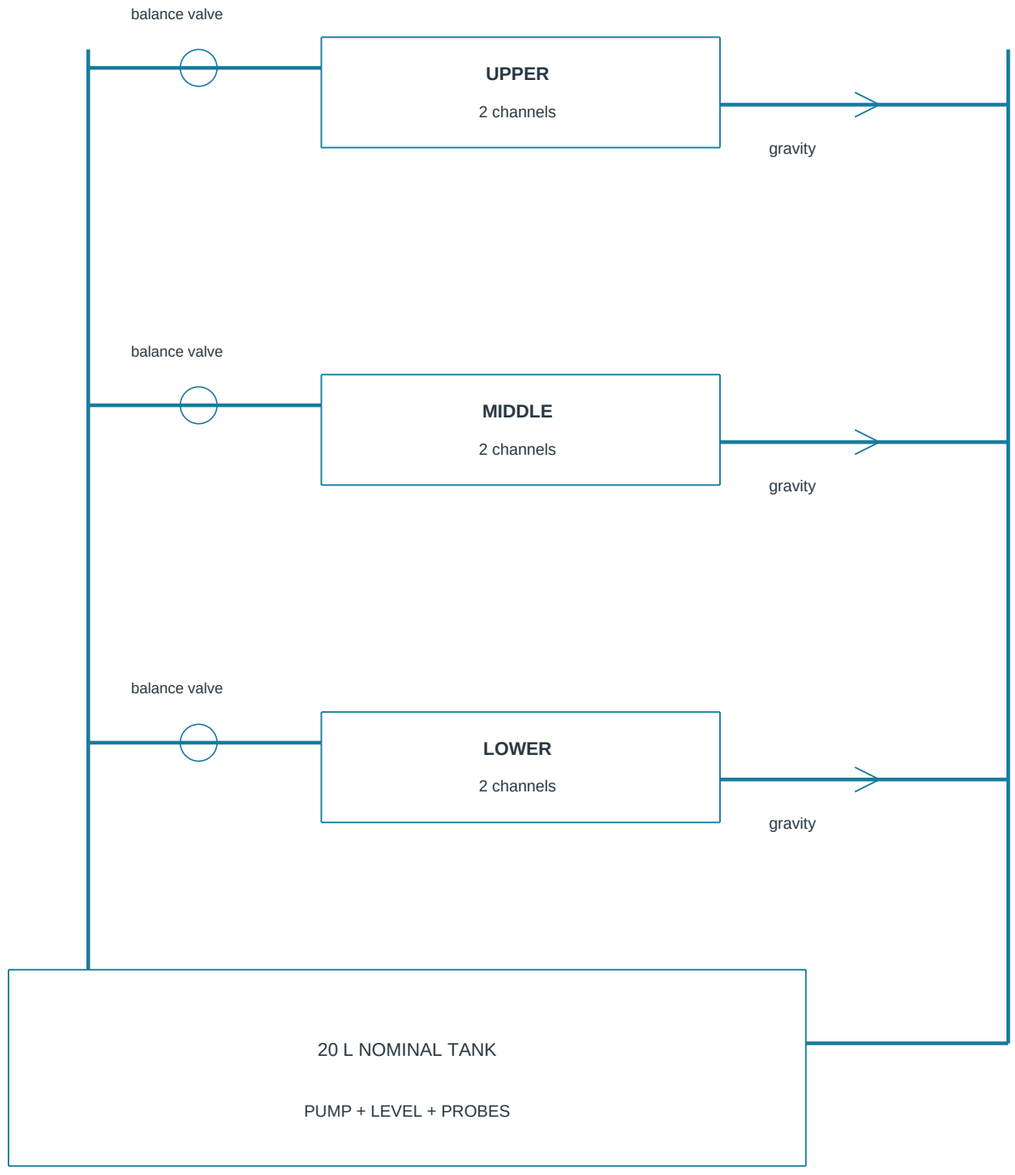
ENGINEERING REVIEW - NOT RELEASED FOR FABRICATION

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Source: SSOT v1.0 + design PDF + video

PARALLEL FEED / GRAVITY RETURN



Topology only: unions, tees, clips and elbows must be detailed after selection.

Pipe and flow proposals - validate physically

Feed riser reserve: 12 OD / 8 ID at X65, Y407, from Z410 to 1650. Tier headers: 16 OD / 12 ID at Y405, Z635 / 1135 / 1635, from X65 to 425. Provide reduction fittings and 6 ID drops to each channel.

Return riser reserve: 40 OD / 32 ID at X60, Y25, from Z410 to 1590. Tier collection headers at Y25, Z480 / 980 / 1480, X60 to 425. Connect both channel outlets with short removable drains clear of tier supports.

Straight routes in STEP are spatial references: missing tees, flexible connectors, valves, slopes at collection headers and unions must be resolved by the plumbing engineer. Keep all fittings within envelope and clear of braces.

Fit balancing/shutoff valves for individual channels. Use a continuous-duty pump; select against actual upper-tier delivery head plus pipe/fitting losses, not zero-head flow. Begin commissioning around 0.2-0.3 L/min per channel, then adjust for uniform wetting and crop/root conditions.

Fault behaviour and service

Normal NFT operation requires sustained nutrient flow; avoid long timer-off periods without crop evidence. Low level or leak response switches pump off and raises an alert. An outage can dry roots, so define backup response before crop trials.

Vent gravity returns; avoid trapped air and unintended siphons. Use retained free-discharge feed ends, accessible root guards and no hidden leaking joints inside structural posts. Add union above tank for removal and a visible return path.

Commission with clean water: balance each channel, stop/restart pump, block one outlet under supervision, and confirm no spill toward electronics. A blocked-channel safeguard and return sizing must be proven before unattended operation.

Basis for hydraulic starting assumptions

Oregon State Extension EM 9457: minimum 2% NFT slope. Virginia Tech SPES-463NP: continuous-duty pumping and 3-5 US gal/h per channel (about 0.19-0.32 L/min). These are initial test inputs, not guaranteed performance of this short-channel design.

H03 SADDLE TOP PLATE - PROPOSED

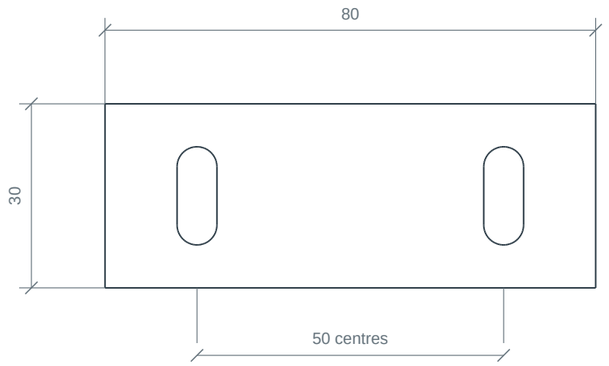
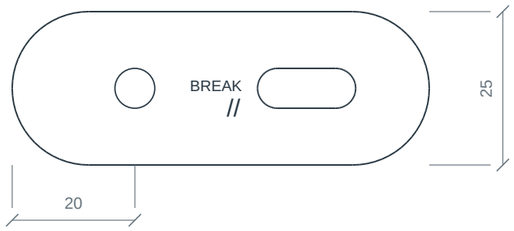


Plate 80 x 30 x 3; 2 slots 6.5 wide x 16 overall, long axis Y. Slot centres X=15/65, Y=15. Slot overall length includes the semicircular ends. Form or shim bearing face to channel slope; isolate with 2 mm pad.

Required total rise at support centres: 20.8 front / 27.2 rear. Includes plate, pad and captive M6 adjuster. Engineer to choose adjuster, locknut and rail attachment. 12 complete saddles required.

BRACE END / STRAP FLAT



2 rear braces: 25 x 2 flat, hole-centre length 1916.7; blank 1956.7. One right brace: hole-centre length 1865.1; blank 1905.1. Overall views broken; use lengths, not sketch scale.

One end dia6.5; opposite end 6.5 x 16 longitudinal slot. 20 nominal centre-to-end; R12.5 rounded ends. Rear straps at Y=414/417 with 1 clear at crossing. Right strap at X=567. Use spacers/offset tabs at joints; verify clashes and minimum edge distances.

Rear brace hole coordinates: X15/585, Z125/1955. Right brace coordinates: Y45/405, Z125/1955. Flat tension bracing is a design proposal; loads and attachment capacity require engineering review.

GENERAL FABRICATION TOLERANCES [P]

Frame saw cuts +/-0.5. Assembled outside size +/-2. Tier support Z +/-1 before slope adjustment. Frame diagonals within 2. Channel transverse level within 0.5; longitudinal fall 8 +/-1. Deburr all cut edges; no sharp accessible corners.

Tolerances are proposed shop targets, not an adopted ISO class. Slot/hole locations for bought-out hardware remain HOLD. Vendor specifications govern fit and tightening torque.

Wall restraint and stability

Two restraint interfaces at upper rear, nominal Z1930 and X30/570. Reserved plates 60 x 60 x 4. Bracket hole pattern and wall anchor are HOLD pending wall substrate and required force; do not drill a generic wall-anchor pattern.

The 2 m high, 450 mm deep frame must not be assumed stable from reservoir weight. Fix restraints before wet or loaded tests. Verify both full-tank and empty-tank cases with the engineer's lateral load and tip criterion.

Proposed load cases for calculation

Design inputs, not load ratings: 25 kg on base including full tank and equipment; 5 kg per grow tier including crop, trapped water and channel; luminaire/support dead loads; measured frame mass. Add foreseeable handling and lateral loads.

Assess member bending, joint slip, local tank support, brace action and restraint pull-out. Proposed service deflection target: <=2 mm at a loaded channel support, with residual set <=0.5 after unloading. Engineer to set proof load before testing.

Material and interface choices

Frame: corrosion-resistant aluminium extrusion, 30 square, exact alloy and section to supplier certification. Isolate dissimilar metal straps/fasteners. No welds assumed. Use cleanable opaque food-contact polymer in wet paths.

Base deck: 3 mm aluminium proposal on perimeter + cross supports. Splash barrier/tray: compatible non-absorbent polymer. No inferred structural capacity or IP rating is certified by this drawing set.

ZOVUYO INNOVATIONS / SMART GROWKIT

Build sequence, procurement schedule and release checklist

ZI-SGK-M010

REV A | 15 SEP 2026

Mechanical procurement and fabrication schedule

ID	Description	Qty	Reference
F01-05	30 x 30 extrusion / 19.8 m net	29	M003
F06	Foot + adaptor, 50 installed height	4	Supplier HOLD
F07-08	Rear X / right tension braces	3	M009
F09	Upper restraint interfaces	2	M009 / HOLD
H01	400 x 100 x 50 NFT bodies	6	M005 / HOLD
H02	400 x 100 x 3 removable lids	6	M005 / HOLD
H03	Adjustable channel saddles	12	M004 / M009
R01	Opaque 20 L nominal reservoir	1	M006 / HOLD
B01	540 x 390 x 3 deck	1	M006
B02	520 x 270 x 50 spill tray	1	M006
B03	20 x 240 x 49 tank bearers	2	M006
E01-02	Control enclosure + splash barrier	1 each	M006 / M007
L01-02	LED envelopes / support strips	6 / 6	M004 / HOLD
V01	Guarded fan envelopes	3	M004 / HOLD
S01-06	Baseline sensors + CO2 module	1 set	M007 / HOLD
P01-05	Pump, feed, return and fittings	1 set	M008 / HOLD

Stock list excludes unselected connector kits, fasteners, cable clips, sealing hardware, tank lid interfaces and wall anchors. Exact quantities and part numbers must be added after supplier selection; it is not a purchase-ready BOM.

Assembly order

1. Confirm component envelopes, connectors, pot depth, electrical budget and wall substrate. Close the holds below before production cuts.
2. Cut and deburr frame; build base/top, posts and tier rails. Fit base crossmembers. Square, brace and restrain frame before loading.
3. Install deck, tray, tank bearers and splash barrier. Check tank removal and electronics access with full-size component mock-ups.
4. Install saddles and channels; set 8 mm fall. Mount LEDs, guarded fans and removable sensor carriers. Protect all cable edges.
5. Fit plumbing and accessible unions; test with clean water before energising integrated electronics. Inspect wiring, sealing and strain relief.
6. Balance flows; calibrate water levels, probes and drain-back reserve. Verify interlocks, restart, offline operation and the seven-day soak.

Release holds - sign and record evidence

H1 Mechanical engineer: section and joint capacity, bracing, deflection, wall-restraint loads and stability evidence.

H2 Builder: measured tank/enclosure/pot/fitting envelopes; all final fixing and penetration drawings; interference review with actual parts.

H3 Hydroponics lead: channel material, slope, outlet/root space, pump duty/head, balancing and drain-back/blocked-return tests.

H4 Electrical lead: final sensor/LED/fan models, power budget, branch protection, dry/wet separation and thermal tests.

H5 Product owner: approve CO2 sensing, 12-site layout, parallel feed and open-frame climate limitations; revise SSOT decision log.

SOURCE REGISTER

S1: Zovuyo Smart GrowKit Source of Truth v1.0, 30 Aug 2026, sections 3/11/14. S2: Design Summary, 5 Feb 2026, pp6-7. S3: supplied 31.53 s screen recording; visual reference only. External hydraulic sources and exact links are in README.

ENGINEERING REVIEW - NOT RELEASED FOR FABRICATION

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